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Prickles

Prickles, are one type of epidermal emergences with no vascular supply. They are stiff and sharp in appearance. (Example: Rose).

Figure 9.17: Prickles

Functions of Epidermal Tissue System

1. This system in the shoot checks excessive loss of water due to the presence of cuticle.
2. Epidermis protects the underlying tissues.
3. Stomata is involved in transpiration and gaseous exchange.
4. Trichomes are also helpful in the dispersal of seeds and fruits, and provide protection against animals.
5. Prickles also provide protection against animals and they also check excessive transpiration
6. In some rose plants they also help in climbing.
7. Glandular hairs repel herbivorous animals.

9.5 Fundamental Tissue System

The ground or fundamental tissue system constitutes the main body of the plants. It includes all the tissues except epidermis and vascular tissues. In monocot stem, ground tissue system is a continuous mass of parenchymatous tissue in which vascular bundles are found scattered. Hence ground tissue is not differentiated into cortex, endodermis, pericycle and pith. Generally in dicot stem, ground tissue system is differentiated into three main zones – cortex, pericycle and pith. It is classified into extrastelar ground tissue (Examples: cortex and endodermis) and intrastelar ground tissue (Examples: pericycle, medullary ray and pith)

Extrastelar Ground Tissue

The ground tissues present outside the stele is called extrastelar ground tissue. (Cortex)

Intrastelar Ground Tissue

The ground tissues present within the stele are called intrastelar ground tissues. (pericycle, medullary rays and pith).

Different Components of Ground Tissue Systems are as follows

Hypodermis

One or two layers of continuous or discontinuous tissue present below the epidermis, is called hypodermis. It is protective in function.

In dicot stem, hypodermis is generally collenchymatous, whereas in monocot stem, it is generally sclerenchymatous. In many plants collenchyma form the hypodermis.

General Cortex

The Cortex occurs between the epidermis and pericycle. Cortex is a few to many layers in thickness, In most cases, it is made up of parenchymatous tissues. Intercellular spaces may or may not be present.

The cortical cells may contain non living inclusions of starch grains, oil, tannins and crystals.

Sometimes in young stem, chloroplasts develop in peripheral cortical cells, which is called **chlorenchyma.**